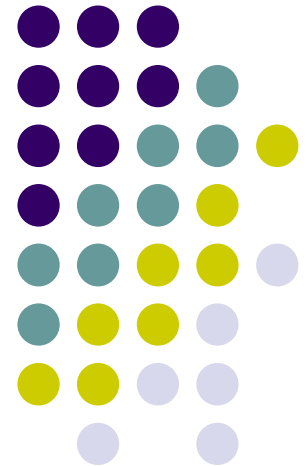
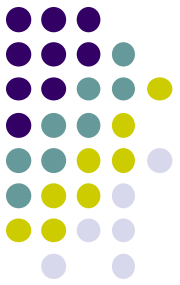


FILTRATION

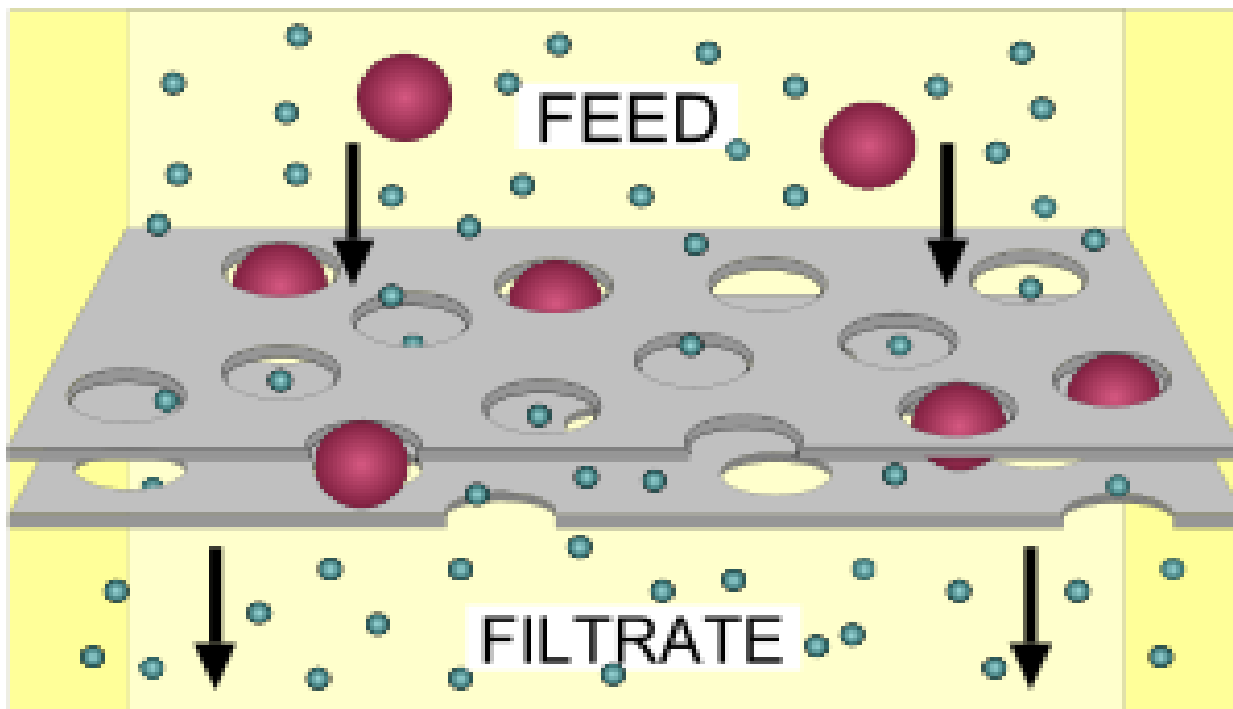


Muluken N.

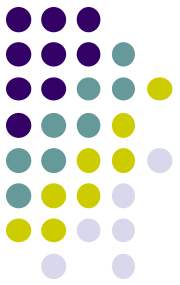


Filtration

- Filtration is the separation of a solid from a fluid by means of a porous medium that retains the solid but allows the fluid to pass.

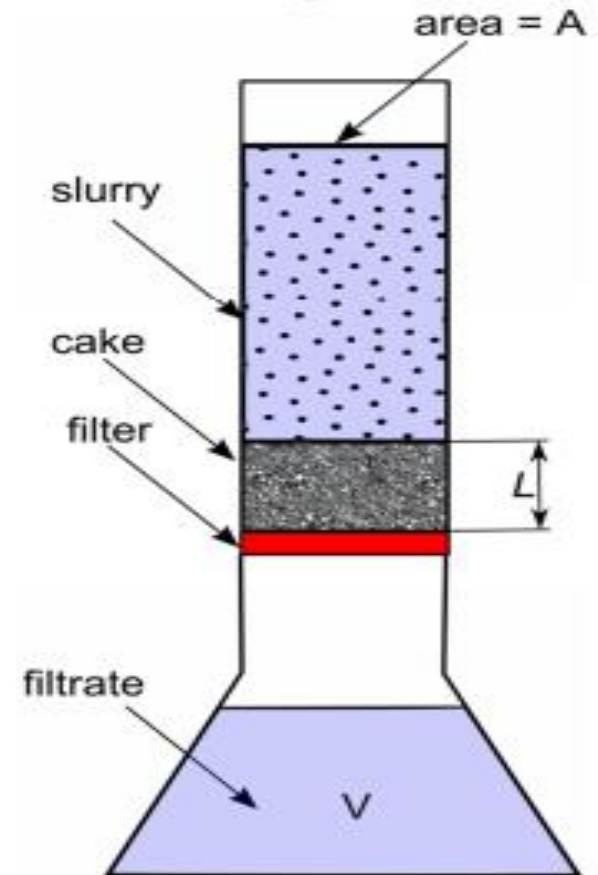


Filtration...

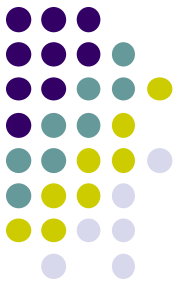


Terms

- **Slurry** - the suspension of solid and liquid to be filtered
- **Filter cake** = the accumulation of solids on the filter
- **Filtrate** = the clear liquid passing through the filter
- **Filter medium** - the porous medium used to retain the solids

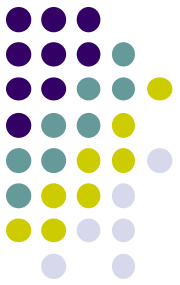


Filtration...

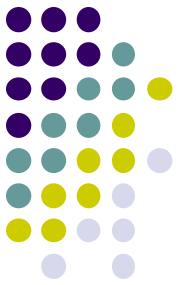


- **Filter medium** can be made from filter paper, woven material as cotton, silk nylon, perforated sheet metal
- Filter medium should be:
 - Inert
 - Strong
 - High retention to solids
 - Low resistance to flow
- **Filter aids-** these are substances when added to liquids to be filtered, reduces the resistance of the filter cake and increases the filtration

Filtration...

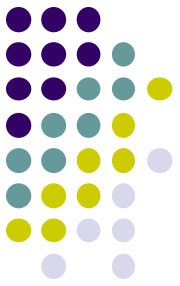


- The objective of the filter aid is to prevent the medium from becoming blocked and to form an **open, porous cake**, so reducing the resistance to flow of the filtrate.
- Example:-
 - **Talc**
 - **Kaolin**
 - **Kieselguhr**



Factors Affecting Filtration

- The unit operation of filtration is affected by the characteristics of slurry which includes:
 - **Properties of liquid**
 - Density
 - Viscosity
 - Corrosiveness
 - **Properties of the solid**
 - Particle size
 - Particle shape
 - Particle size distribution
 - Compressibility of the solid
 - **The proportion of solid in the slurry**



Factors Affecting Rate of Filtration

The factors which govern the rate of filtration are:

A. Area of filter surface

- Rate is directly proportional to area

B. Particle size of the solid to be removed

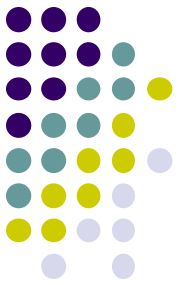
- Directly proportional

C. Pore size of filter medium

- Directly proportional

D. Viscosity of the liquid to be filtered

- Inversely proportional
- Liquids with low viscosity gets filtered quickly and vice versa



Factors Affecting Rate...

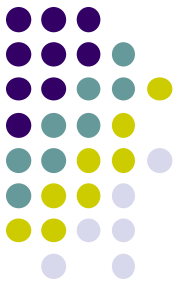
E. Temperature

- Increased T° reduces the viscosity of thick liquids and can be filtered quickly

F. Pressure difference across the filter medium

- Rate can be increased by increasing the pressure on the liquid to be filtered or by reducing the pressure under the filter medium

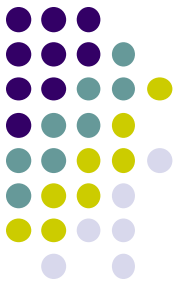
Methods and Equipments used for Filtration



Equipment selection

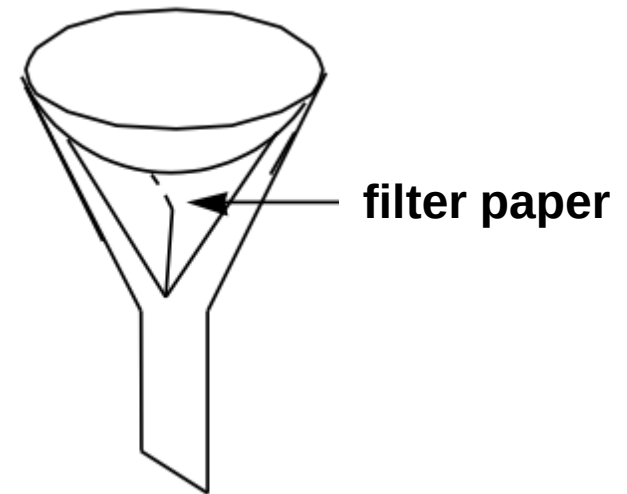
- ❖ Ideally the equipment chosen should allow a fast filtration rate to minimize production costs:
 - ✓ be cheap to buy and run,
 - ✓ be easily cleaned and resistant to corrosion, and
 - ✓ be capable of filtering large volumes of product before the filter needs stripping down for cleaning or replacing

Methods and Equipments...



1. Filter funnels

- These are conical shaped devices used for filtration
- Can be made from glass, copper, plastic or stainless steel

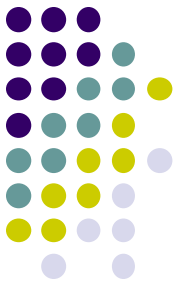


2. Buckner funnel

- Used for filtration under reduced pressure

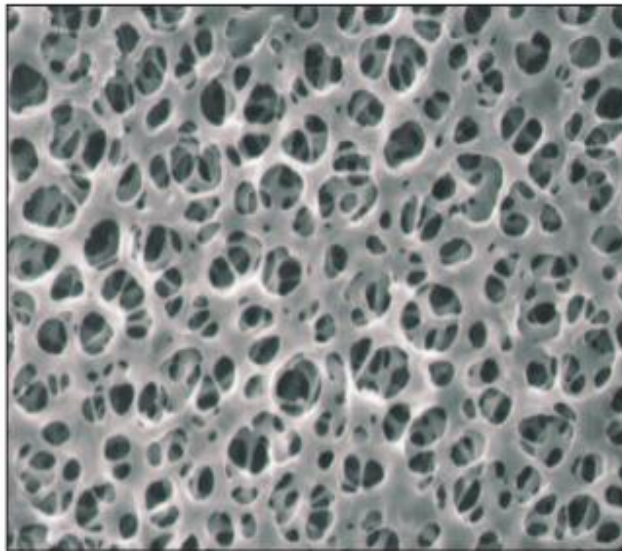


Methods and Equipments...

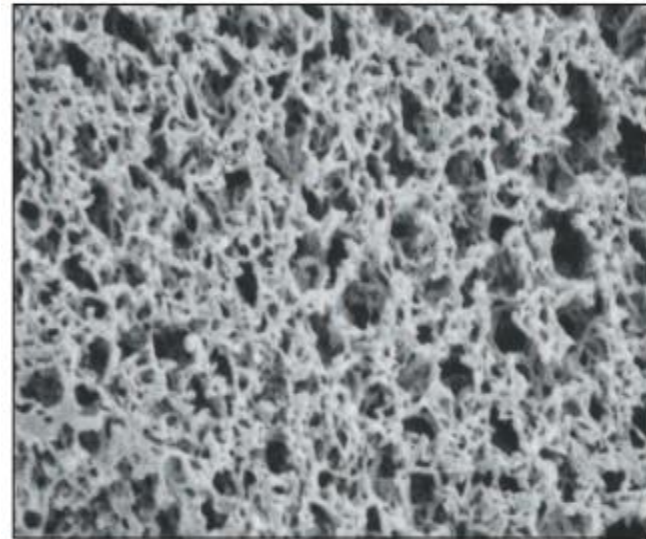


3. Bacteria proof filter candles

- Membrane filter
 - Made of cellulose, PVC, nylon
 - Are very fine having a pore size of 0.22 - 0.45 micron

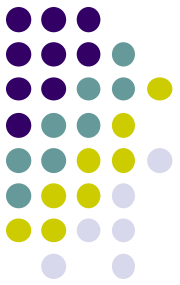


Cellulose Acetate



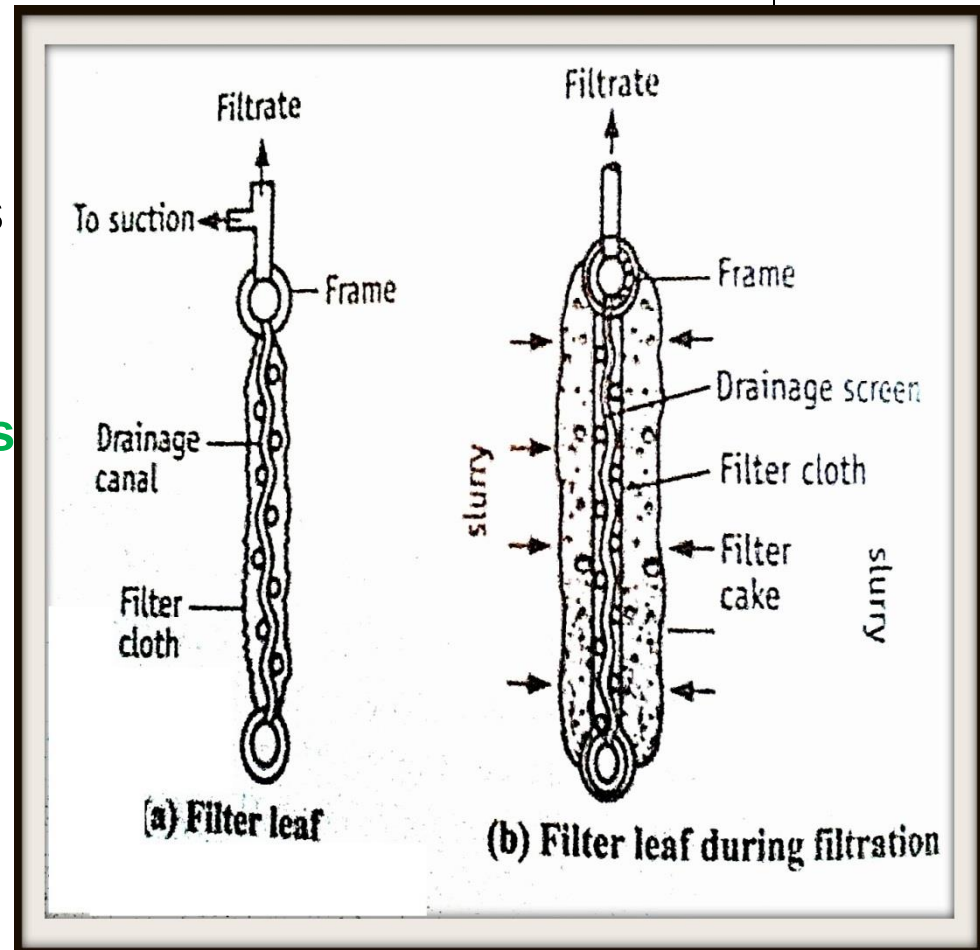
Nylon

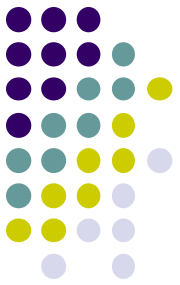
Methods and Equipments...



4. Filter leaf

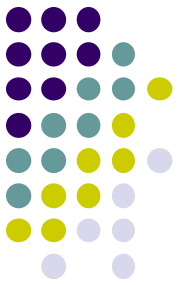
- It consists of a frame in which a drainage screen is enclosed
- The whole unit is covered with a **cloth which acts as a filter medium**
- The inside of the frame is fitted with an outlet and an attachment to connect to **vacuum pump**





Methods and Equipments...

- For use the filter leaf is immersed in the container having the slurry to be filtered. It is then connected to vacuum pump
- When vacuum is created, filtration will start and the cake will be deposited on the outer surface of filter cloth
- Use for the filtration of slurry which do not contain high solid content, about **5%**, i.e. dilute suspensions.

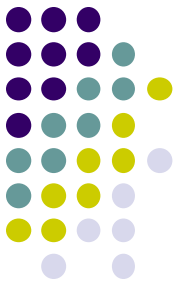


Methods and Equipments...

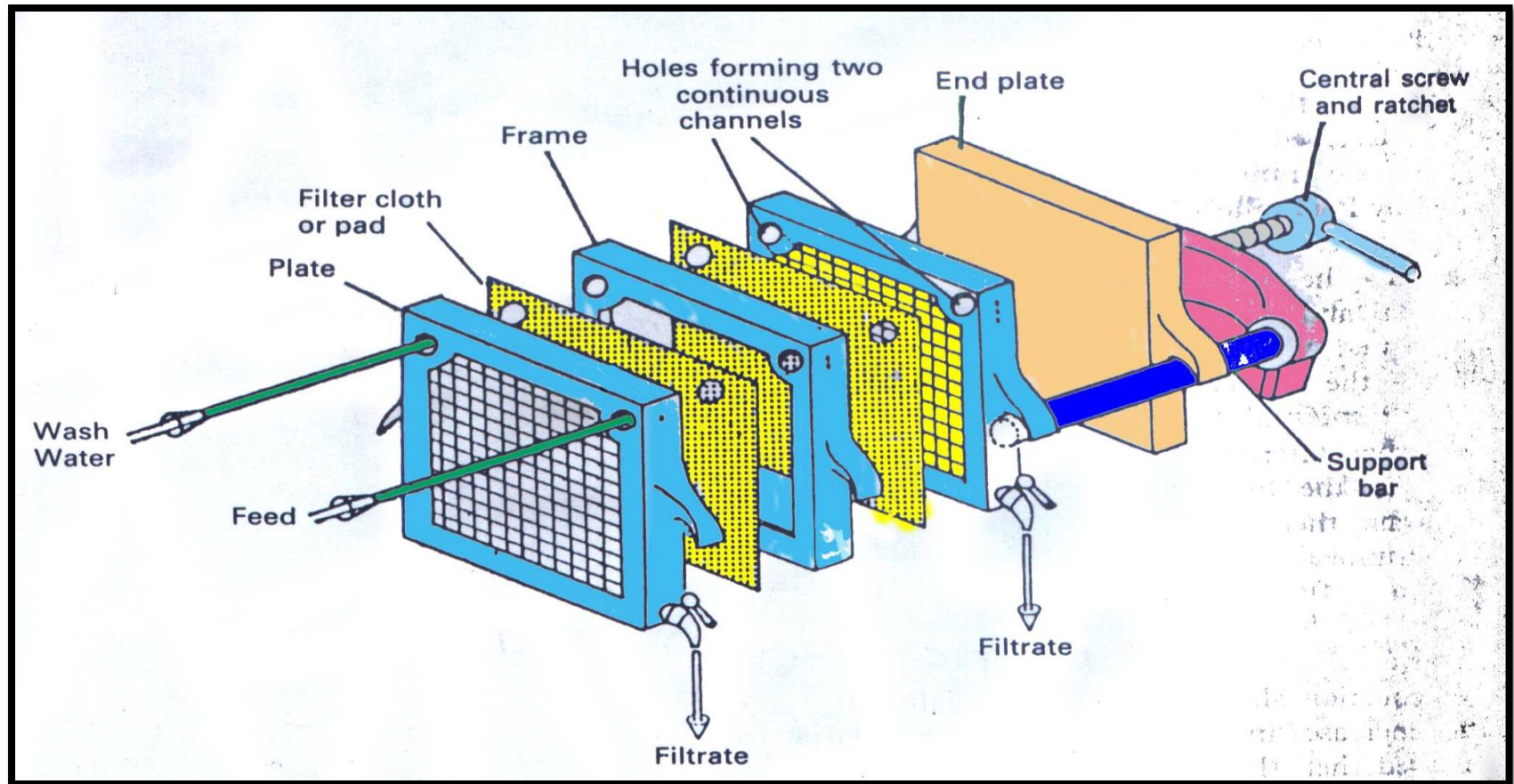
5. Filter press

- It consists of a number of hollow **frames** and solid **plates** that support filter media
- The plates and frames are assemble alternatively and each set works as a unit
- The liquid to be filtered is allowed to enter the frame under pressure
- The filtrate passes through the filter medium on the surface of the plate while the filter cake is retained in the frame

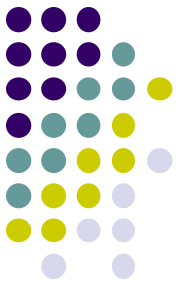
Methods and Equipments...



Filter press

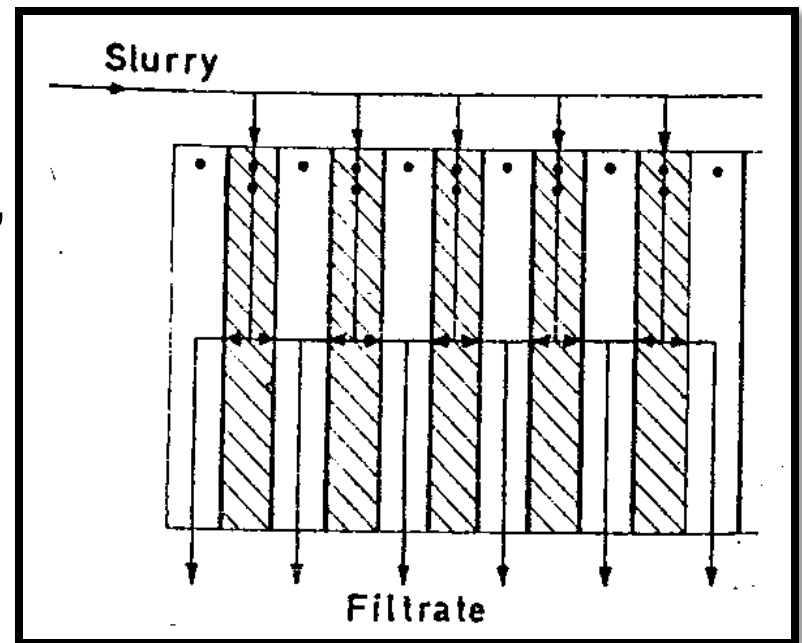


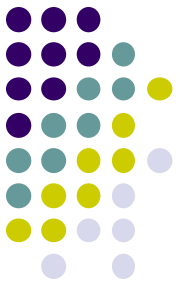
Methods and Equipments...



Filter press working principle

- Slurry enters the frame from the feed channel and passes through the filter medium on the surface of the plate
- The solid forms a filter cake and remain in the frame
- Thus two filter cakes are formed, which meet eventually in the centre of the frame
- The filtrate drains between the projections of the surface of the plate and escape from the outlet

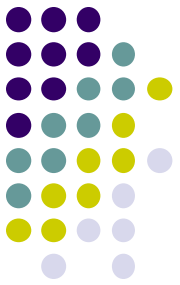




Methods and Equipments...

Advantages

- Provide large filtration area in relatively small floor space
- Sturdy construction permits the use of considerable pressure difference (2000 Kilopascals normally used)
- Efficient washing of cake is possible
- Operation and maintenance is easy
- It produce dry cake in form of slab

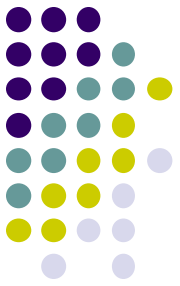


Methods and Equipments...

Disadvantages

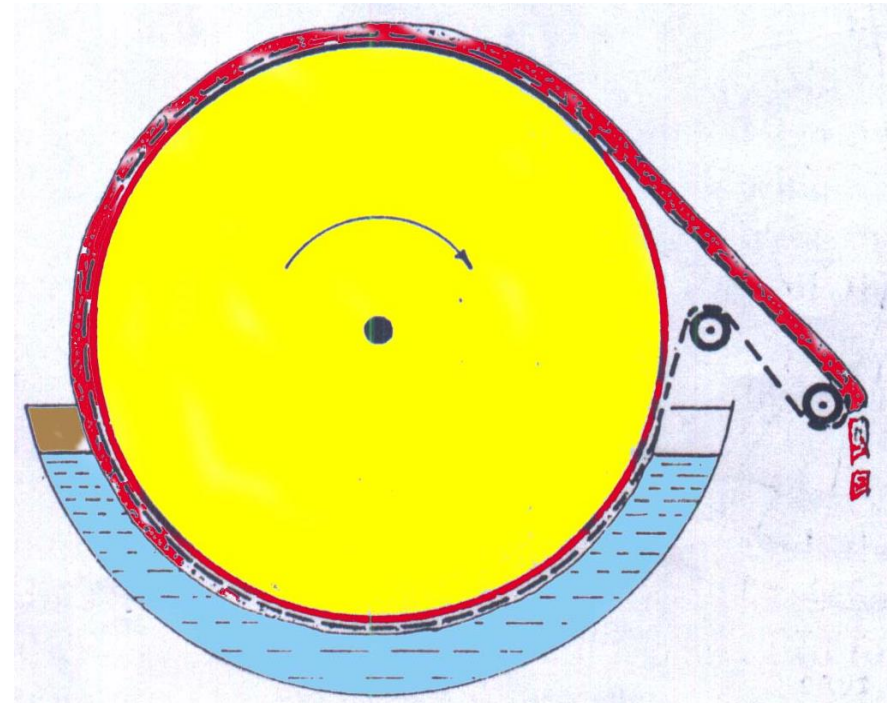
- It is a batch filter, so it is a time consuming.
- The filter press is an expensive filter, the emptying time, the labor involved, and the wear and tear on the cloths resulting in high costs.
- Operation is critical, as the frames should be full, otherwise washing is inefficient.
- The filter press is used for slurries containing less about **5 %** solids

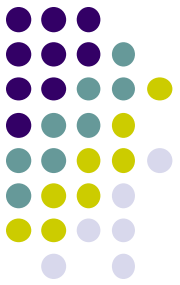
Methods and Equipments...



6. Rotary filter

- It consists of a metal drum mounted horizontally.
- The drum may be up to 3 meters in diameter and 3.5 meters in length
- The curved surface being a perforated plate, supporting a filter cloth.
- Slurry filtered through sieve like mechanism on the rotation drum surface, under the condition of vacuum.





Methods and Equipments...

- Internally, it is divided into several sectors and a separate connection is made between each sector and a special rotary valve.
- The drum is dipped into the slurry and vacuum applied to the outlet, which is connected to the filtrate receiver.
- The drum is sprayed with water to wash the cake.
- It is used when the solid content is in the range of **15-30%**
- *It is continuous process*
- Used for collection of CaCO_3 , MgCO_3 etc